



Luca Lazzaroni

Ricercatore a tempo determinato in Elettronica

DITEN — Dip. di Ingegneria Navale, Elettrica, Elettronica e delle Telecomunicazioni

SEDE Genova, Italia

EMAIL luca.lazzaroni@unige.it

SETTORE RTD-A · IINF-01/A

AGGIORNATO In ruolo da marzo 2025

01 Profilo

Ricercatore a tempo determinato (RTD-A) presso il DITEN dell'Università degli Studi di Genova, settore IINF-01/A — Elettronica, dal marzo 2025. Dottore di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni, 2024.

Attività di ricerca su machine learning embedded ed edge AI: toolchain TinyML, reti neurali quantizzate e binarie su microcontrollori e FPGA, percezione e decisione per la guida automatizzata, deep reinforcement learning, assistenti vocali on-device. L'attività si svolge nell'ambito di progetti di ricerca europei e nazionali con partner automotive e industriali.

Edge AI

TinyML

Deep learning embedded

Guida automatizzata

Deep reinforcement learning

Assistenti vocali embedded

02 Incarichi e formazione

POSIZIONI

Mar 2025 — oggi

Ricercatore a tempo determinato (RTD-A)

DITEN, Università di Genova

Settore scientifico-disciplinare IINF-01/A — Elettronica.

Nov 2023 — Feb 2025

Assegnista di ricerca

DITEN, Università di Genova

Assegno: “Realizzazione di modelli di Machine Learning per il trattamento di dati provenienti da veicoli autonomi”.

Nov 2020 — Ott 2023

Dottorando di ricerca

DITEN, Università di Genova — XXXVI ciclo

Borsa di dottorato: “Deep learning on the edge”.

FORMAZIONE

2024

Dottorato di ricerca in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni

Università di Genova — conseguito il 14 marzo 2024

Tesi: “The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning”

2020 **Laurea Magistrale in Ingegneria Elettronica (LM-29)**
Università di Genova — 24 luglio 2020
Tesi: “Progetto e implementazione di un motore di esecuzione multiplatforma per applicazioni IoT” · Voto: 110/110

2017 **Laurea in Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)**
Università di Genova — 27 ottobre 2017
Tesi: “Progetto model-based del controllo di robot Lego in ambiente Simulink”

2014 **Diploma di Maturità Scientifica**
Liceo Scientifico Statale “Arturo Issel”, Finale Ligure (SV)
Voto: 100/100

Abilitazione all'esercizio della professione di Ingegnere dell'Informazione — Esame di Stato superato nella seconda sessione 2020.

03 Bibliometria

AGGIORNATO IL 17 SET 2026

66 DOCUMENTI SCOPUS	336 CITAZIONI	11 INDICE H	71 CO-AUTORI
-------------------------------	-------------------------	-----------------------	------------------------

Tutti i valori provengono da Scopus (author ID 57220892898) e sono aggiornati automaticamente. Le 6 voci di convegni nazionali e workshop non indicizzate da Scopus sono elencate sotto, ma non concorrono a nessuno di essi.

04 Pubblicazioni

72 VOCI · 66 SU SCOPUS

2026 **Cross-platform deployment and characterization of an on-device generative conversational agent**
P. Firpo, R. Berta, F. Bellotti, **L. Lazzaroni**, M. Fresta, A. Pighetti, A. Abedinia, N. Yoonessi, D.P. Pau
Journal of Systems Architecture vol. 181, art. 103981 Rivista
<https://doi.org/10.1016/j.sysarc.2026.103981>

A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction
L. Forneris, R. Berta, M. Fresta, **L. Lazzaroni**, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, F. Bellotti
IEEE Signal Processing Letters vol. 33, pp. 136–140 Rivista
<https://doi.org/10.1109/lsp.2025.3638676>

FPGA Deployment of Extremely Quantized Models for Time-to-Lane-Change Prediction
A. Saad, R. Berta, M. Fresta, **L. Lazzaroni**, A. Pighetti, O. Srouf, F. Tango, F. Bellotti
IEEE Transactions on Circuits and Systems II: Express Briefs vol. 73, no. 8, pp. 888–892 Rivista
<https://doi.org/10.1109/tcsii.2026.3705103>

Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms
H. Rojhan, F. Bellotti, M. Fresta, **L. Lazzaroni**, R. Berta
IEEE Access vol. 14, pp. 44886–44903 Rivista
<https://doi.org/10.1109/access.2026.3676567>

StabilEdge, Formal Verification of Edge AI Applications

L. Lazzaroni, R. Berta, V. Soltanmuradov, D. Martin Gomez, F. Bellotti

IEEE Access vol. 14, pp. 37416–37434 Rivista

<https://doi.org/10.1109/access.2026.3667991>

Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers

A. Saad, F. Bellotti, H. Ballout, M. Fresta, L. Lazzaroni, O. Srour, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 321–327 Conferenza

https://doi.org/10.1007/978-3-032-17174-0_46

Preface

M. Ruo Roch, R. Berta, A. Gómez, L. Lazzaroni, M. Martina, P.M. Ros, N. Pazos Escudero

Lecture Notes in Electrical Engineering pp. v–vi Editoriale

Optimizing Binary AutoEncoders on STM32 Microcontrollers

H. Ballout, R. Berta, L. Lazzaroni, F. Sakr, A. Pighetti, M. Fresta, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 301–307 Conferenza

https://doi.org/10.1007/978-3-032-17174-0_43

Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students

L. Motta, F. Bellotti, R. Berta, L. Lazzaroni, M. Fresta, N. Busi

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–248 Conferenza

https://doi.org/10.1007/978-3-032-17174-0_34

Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform

H. Rojhan, F. Bellotti, L. Lazzaroni, M. Fresta, L. Forneris, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 218–223 Conferenza

https://doi.org/10.1007/978-3-032-17174-0_30

Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA

D. Faryar, R. Berta, M. Fresta, A. Saad, L. Lazzaroni, H. Ballout, O. Srour, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 25–31 Conferenza

https://doi.org/10.1007/978-3-032-17174-0_4

Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning

O.L. Audi, F. Bellotti, R. Berta, Z. Liu, L. Lazzaroni, A. Pilla, H. Rojhan, L. Forneris

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 212–217 Conferenza

https://doi.org/10.1007/978-3-032-17174-0_29

Implementing YoloV8 on Zynq UltraScale + MPSoC FPGA Boards

H. Ballout, R. Berta, O. Srour, L. Lazzaroni, M. Fresta, A. Saad, F. Bellotti

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 289–294 Nazionale

https://doi.org/10.1007/978-3-032-17305-8_31

FPGA Implementation of a Robust Decision-Making System for Low-Speed Maneuvering in Automated Driving

A. Saad, H. Ballout, F. Bellotti, M. Fresta, L. Lazzaroni, O. Srour, R. Berta

Proceedings of SIE 2025 Lecture Notes in Electrical Engineering pp. 295–301 Nazionale

https://doi.org/10.1007/978-3-032-17305-8_32

Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision

A. Pighetti, F. Bellotti, S. Mohammady, **L. Lazzaroni**, H. Ballout, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 85–90

Conferenza

https://doi.org/10.1007/978-3-032-17174-0_12

Embedded Deployment of an LLM-Based Voice Assistant

L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srouf, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 273–278

Conferenza

https://doi.org/10.1007/978-3-032-17174-0_39

Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection

V. Soltanmuradov, F. Bellotti, R. Berta, D.M. Gómez, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 286–291

Conferenza

https://doi.org/10.1007/978-3-032-17174-0_41

A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors

S.G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, **L. Lazzaroni**, H. Ballout, H. Rojhan, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 183–190

Conferenza

https://doi.org/10.1007/978-3-032-17174-0_25

A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems

E. Castagno, F. Bellotti, R. Berta, **L. Lazzaroni**, M. Fresta, S. Incerti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 249–254

Conferenza

https://doi.org/10.1007/978-3-032-17174-0_35

A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents

A. Zare, F. Bellotti, R. Berta, **L. Lazzaroni**, L. Forneris, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 79–84

Conferenza

https://doi.org/10.1007/978-3-032-17174-0_11

2025

An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection

L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, R. Berta

IEEE Sensors Journal vol. 25, no. 24, pp. 44859–44874

Rivista

<https://doi.org/10.1109/jsen.2025.3627477>

TinyML Acceleration with MAX78000

A. Dabbous, **L. Lazzaroni**, F. Bellotti, S.M. Presentación, A. Pighetti, R. Berta

Proceedings of SIE 2024 Lecture Notes in Electrical Engineering pp. 468–474

Nazionale

https://doi.org/10.1007/978-3-031-71518-1_57

Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection

L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 156–163

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_19

Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking

H. Ballout, R. Berta, I. Bochenko, **L. Lazzaroni**, A. Dhonthi, V. Hashemi, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 387–394

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_46

Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking

F. Bellotti, R. Berta, V. Soltanmuradov, D.M. Gómez, A. Dhonthi, V. Hashemi, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 139–147

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_17

RL-Based Generation of a Synthetic Automotive Driving Scenario Dataset

R. Berta, F. Bellotti, M. Cossu, L. Forneris, **L. Lazzaroni**, A. Pighetti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 395–402

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_47

RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

IEEE Sensors Letters vol. 9, no. 9, art. 6010204

Rivista

<https://doi.org/10.1109/lensens.2025.3600982>

Quantization of MobileNetV2 for Resource-Constrained Microcontrollers

L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 497–504

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_59

Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board

H. Ballout, R. Berta, A. Dabbous, M. Fresta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 242–249

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_29

On-Board Deployability of a Deep Learning-Based System for Distraction and Inattention Detection

M. Fresta, F. Bellotti, **L. Lazzaroni**, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, R. Berta

IEEE Open Journal of the Industrial Electronics Society vol. 6, pp. 1351–1369

Rivista

<https://doi.org/10.1109/ojies.2025.3601982>

Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices

A. Capello, R. Berta, M. Fresta, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 323–330

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_38

Green Machine Learning for the IoT (GEMINI): A Position Paper

L. Lazzaroni, A. Dabbous, M. Nardello, M.A. Lebdeh, R. Berta, K.S. Yildirim, F. Bellotti, D. Brunelli

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 339–346

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_40

Exploring Decision Transformer for Highway Automated Driving

L. Forneris, F. Bellotti, R. Berta, **L. Lazzaroni**, C. Oh

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 123–130

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_15

Deep Learning-Based Real-Time Driver Cognitive Distraction Detection

M. Fresta, F. Bellotti, I. Bochenko, **L. Lazzaroni**, G. Merlhiot, F. Tango, R. Berta

IEEE Access vol. 13, pp. 26589–26607

Rivista

<https://doi.org/10.1109/access.2025.3539392>

Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection

M. Cossu, R. Berta, M.A. Bagheri Orumi, **L. Lazzaroni**, J.L. Sauvaget, H. Touil, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 250–257

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_30

Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices

A. Dabbous, **L. Lazzaroni**, F. Sakr, R. Berta, M. Fresta, F. Bellotti

2025 IEEE International Conference on Consumer Electronics (ICCE)

Digest of Technical Papers IEEE International Conference on Consumer Electronics

Conferenza

<https://doi.org/10.1109/icce63647.2025.10930047>

Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification

A. Dabbous, R. Berta, **L. Lazzaroni**, A. Pighetti, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 234–241

Conferenza

https://doi.org/10.1007/978-3-031-84100-2_28

2024

An embedded end-to-end voice assistant

L. Lazzaroni, F. Bellotti, R. Berta

Engineering Applications of Artificial Intelligence vol. 136, art. 108998

Rivista

<https://doi.org/10.1016/j.engappai.2024.108998>

Adversarial Policy Generation in Automated Parking

A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, **L. Lazzaroni**, C. Oh

ECCV 2024 Workshop on Multimodal Perception and Comprehension of Corner Cases in Autonomous Driving

Workshop

CBin-NN: An Inference Engine for Binarized Neural Networks

F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Bellotti

Electronics vol. 13, no. 9, art. 1624

Rivista

<https://doi.org/10.3390/electronics13091624>

End-to-End Dataset Collection System for Sport Activities

M. Fresta, F. Bellotti, A. Capello, A. Dabbous, **L. Lazzaroni**, F. Ansovini, R. Berta

Electronics vol. 13, no. 7, art. 1286

Rivista

<https://doi.org/10.3390/electronics13071286>

YoloP-Based Pre-processing for Driving Scenario Detection

M. Cossu, R. Berta, L. Forneris, M. Fresta, **L. Lazzaroni**, J.L. Sauvaget, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 418–423

Conferenza

https://doi.org/10.1007/978-3-031-48121-5_60

Low-Cost, Edge-Cloud, End-to-End System Architecture for Human Activity Data Collection

M. Fresta, A. Dabbous, F. Bellotti, A. Capello, **L. Lazzaroni**, A. Pighetti, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 444–449

Conferenza

https://doi.org/10.1007/978-3-031-48121-5_64

Investigating Adversarial Policy Learning for Robust Agents in Automated Driving Highway Simulations

A. Pighetti, F. Bellotti, C. Oh, **L. Lazzaroni**, L. Forneris, M. Fresta, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 124–129

Conferenza

https://doi.org/10.1007/978-3-031-48121-5_18

Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity

R. Berta, **L. Lazzaroni**, A. Capello, M. Cossu, L. Forneris, A. Pighetti, F. Bellotti

Journal of Intelligent and Connected Vehicles vol. 7, no. 3, pp. 229–244

Rivista

<https://doi.org/10.26599/jicv.2023.9210039>

Developing a TinyML Image Classifier in an Hour

R. Berta, A. Dabbous, **L. Lazzaroni**, D.P. Pau, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 946–960

Rivista

<https://doi.org/10.1109/ojies.2024.3451959>

Building a Pipeline for Efficient Production of Synthetic Datasets for Improving RL in Automated Driving

L. Lazzaroni, A. Pighetti, F. Bellotti, R. Berta

Proceedings of SIE 2023 Lecture Notes in Electrical Engineering pp. 351–356

Nazionale

https://doi.org/10.1007/978-3-031-48711-8_42

Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge

A. Dabbous, R. Berta, M. Fresta, H. Ballout, **L. Lazzaroni**, F. Bellotti

IEEE Open Journal of the Industrial Electronics Society vol. 5, pp. 781–794

Rivista

<https://doi.org/10.1109/ojies.2024.3434341>

Automated Parking in CARLA: A Deep Reinforcement Learning-Based Approach

L. Lazzaroni, A. Pighetti, F. Bellotti, A. Capello, M. Cossu, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 352–357

Conferenza

https://doi.org/10.1007/978-3-031-48121-5_50

Application of a Serious Game for Emotion Elicitation Under Socio-Economic and Trust Based Decision-Making Scenarios for Autistic Adolescents

F. Ahmed, R. Berta, F. Bellotti, F. Floris, **L. Lazzaroni**, G. Barresi, J.R. Carrion

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 170–180

Conferenza

https://doi.org/10.1007/978-3-031-49065-1_17

2023

Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles

L. Forneris, A. Pighetti, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, R. Berta

International Journal of Serious Games vol. 10, no. 4, pp. 153–170

Rivista

<https://doi.org/10.17083/ijsg.v10i4.638>

Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild

F. Ahmed, R. Berta, F. Bellotti, **L. Lazzaroni**, F. Floris, G. Barresi, J.R. Carrion

Applied Sciences vol. 13, no. 11, art. 6432

Rivista

<https://doi.org/10.3390/app13116432>

Supporting a .csv-based Workflow in MongoDB for Data Analysts

M. Fresta, A. Capello, F. Bellotti, **L. Lazzaroni**, M. Cossu, R. Berta

2023 IEEE 32nd International Symposium on Industrial Electronics (ISIE)

IEEE International Symposium on Industrial Electronics

Conferenza

<https://doi.org/10.1109/isie51358.2023.10228044>

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 307–311

Conferenza

https://doi.org/10.1007/978-3-031-30333-3_41

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

IEEE Access vol. 11, pp. 28522–28550

Rivista

<https://doi.org/10.1109/access.2023.3259544>

Efficient Uploading of.Csv Datasets into a Non-Relational Database Management System

M. Fresta, F. Bellotti, A. Capello, M. Cossu, **L. Lazzaroni**, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 9–15

Conferenza

https://doi.org/10.1007/978-3-031-30333-3_2

Developing a Toolchain for Synthetic Driving Scenario Datasets

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 222–228

Conferenza

https://doi.org/10.1007/978-3-031-30333-3_29

Designing an Interpretability Analysis Framework for Deep Reinforcement Learning (DRL) Agents in Highway Automated Driving Simulation

F. Bellotti, **L. Lazzaroni**, A. Capello, M. Cossu, A. De Gloria, R. Berta

Proceedings of SIE 2022 Lecture Notes in Electrical Engineering pp. 239–244

Nazionale

https://doi.org/10.1007/978-3-031-26066-7_37

Deep Reinforcement Learning for Automated Car Parking

L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 125–130

Conferenza

https://doi.org/10.1007/978-3-031-30333-3_16

2022

Synthetic Driving Scenarios Generator

M. Cossu, R. Berta, A. Capello, A. De Gloria, **L. Lazzaroni**, F. Bellotti

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

Nazionale

Investigating High-Level Decision Making for Automated Driving

A. Capello, L. Forneris, A. Pighetti, F. Bellotti, **L. Lazzaroni**, M. Cossu, A. De Gloria, R. Berta

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

Nazionale

Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations

F. Bellotti, R. Berta, A. De Gloria, A. Capello, M. Cossu, **L. Lazzaroni**

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

Nazionale

An Embedded End-to-End Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

53rd Annual Meeting of the Italian Society of Electronics (SIE), Pizzo (VV)

Nazionale

Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications

R. Berta, F. Bellotti, A. De Gloria, **L. Lazzaroni**

Sensors vol. 22, no. 3, art. 713

Rivista

<https://doi.org/10.3390/s22030713>

High-Level Decision-Making Non-player Vehicles

A. Pighetti, L. Forneris, **L. Lazzaroni**, F. Bellotti, A. Capello, M. Cossu, A. De Gloria, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 223–233

Conferenza

https://doi.org/10.1007/978-3-031-22124-8_22

Developing a Synthetic Dataset for Driving Scenarios

J. Motta, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, S. Bonora

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 310–316

Conferenza

https://doi.org/10.1007/978-3-030-95498-7_43

Classifying Simulated Driving Scenarios from Automated Cars

M. Cossu, J.L.Q. Villon, F. Bellotti, A. Capello, A. De Gloria, **L. Lazzaroni**, R. Berta

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 229–235

Conferenza

https://doi.org/10.1007/978-3-030-95498-7_32

A Serious Game for Using Socio-Economic and Trust Based Decision-Making Scenarios for Elicitation of Emotional Responses

F. Ahmed, J.R. Carrion, F. Bellotti, **L. Lazzaroni**, G. Barresi, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 43–52

Conferenza

https://doi.org/10.1007/978-3-031-22124-8_5

2021

Project and Development of an Italian Offline Embedded Voice Assistant

L. Lazzaroni, F. Bellotti, A. De Gloria, R. Berta

52nd Annual Meeting of the Italian Society of Electronics (SIE), Trieste

Nazionale

Edgine, A Runtime System for IoT Edge Applications

R. Berta, A. Mazzara, F. Bellotti, A. De Gloria, **L. Lazzaroni**

Applications in Electronics Pervading Industry, Environment and Society Lecture Notes in Electrical Engineering

pp. 261–266

Conferenza

https://doi.org/10.1007/978-3-030-66729-0_31

Adapting Autonomous Agents for Automotive Driving Games

G. Campodonico, F. Bellotti, R. Berta, A. Capello, M. Cossu, A. De Gloria, **L. Lazzaroni**, T. Taccioli, F. Davio

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 101–110

Conferenza

https://doi.org/10.1007/978-3-030-92182-8_10

2020

Employing an IoT Framework as a Generic Serious Games Analytics Engine

L. Lazzaroni, A. Mazzara, F. Bellotti, A. De Gloria, R. Berta

Games and Learning Alliance

Lecture Notes in Computer Science Including Subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics

pp. 79–88

Conferenza

https://doi.org/10.1007/978-3-030-63464-3_8

05 Progetti di ricerca

Synergies

Generation of scenarios for development, training, virtual testing and validation of CCAM systems

Europeo — HORIZON-CL5-2023-D6-01-02 · Membro dell'unità di ricerca dell'Università di Genova

Genera scenari di test per addestramento, test e validazione di sistemi CCAM, con focus su traffico urbano ed extraurbano — dove la conoscenza degli scenari rilevanti è molto più scarsa che in autostrada. Coinvolge le principali case automobilistiche europee e molte università del settore.

Hi-Drive

Addressing challenges toward the deployment of higher automation

Europeo — H2020, GA 101006664 · Membro dell'unità di ricerca dell'Università di Genova

Sviluppa e testa su larga scala funzionalità di guida automatizzata avanzata, passando dall'automazione condizionata (SAE L3) a quella alta, in diversi scenari e contesti stradali.

GEMINI

Green Machine Learning for the IoT

Nazionale — PRIN 2022 · Membro dell'unità di ricerca dell'Università di Genova (DII–UNITN / DITEN–UNIGE)

Un framework per applicazioni TinyML pienamente sostenibili distribuite lungo un continuum edge-to-cloud, pensato per dispositivi embedded senza batteria, inclusi quelli privi di connettività di rete.

ADINA

Assistente Digitale Integrato Automotive

Regionale — POR FESR Piemonte 2014-2020 · Membro dell'unità di ricerca dell'Università di Genova, con Lifetouch

Un assistente virtuale integrato su hardware dedicato a bordo vettura e connesso ai sistemi del veicolo: algoritmi di IA interpretano le richieste del guidatore e agiscono su funzioni veicolo e infotainment.

06 Didattica e supervisione

A.A. 2024/25, 2025/26	Distributed Electronic Systems and Technologies for Environmental Monitoring Docente titolare · cod. 94631
A.A. 2024/25, 2025/26	Tiny Machine Learning Docente titolare · cod. 114752
dall 'A.A. 2022/23	Machine Learning for Automated Driving Cultore della materia · cod. 108716
A.A. 2023/24	Computer Graphics Arch. — Games and Simulation Docente a contratto · cod. 172317
A.A. 2022/23	Autonomous Agents in Games Supporto alla didattica · cod. 198216
A.A. 2020/21	Fondamenti di Programmazione di Sistemi Elettronici Supporto alla didattica · cod. 166061

Tutti i corsi presso l'Università di Genova. Metodologie didattiche innovative sono state adottate anche per “Fondamenti di Programmazione di Sistemi Elettronici” (166061) e “Approccio Makers alla Progettazione Elettronica” (186985), A.A. 2020/21.

SUPERVISIONE

12	Tesi di Laurea — Ingegneria Elettronica e Tecnologie dell'Informazione (L-8)	Co-supervisore di Ali Zare, Corso di Dottorato in Scienze e Tecnologie per l'Ingegneria Elettronica e delle Telecomunicazioni (XLI ciclo).
10	Tesi di Laurea Magistrale — Ingegneria Elettronica (LM-29)	
7	Tesi di Laurea Magistrale — Ingegneria Tecnologica per la Strategia e la Sicurezza (LM/DS)	Membro della Commissione Didattica e Ammissione del Corso di Laurea Magistrale in Engineering for Natural Risk Management.
2	Tesi di Laurea Magistrale — Informatica (LM-18)	

07 Curatela e servizio

RUOLI EDITORIALI

2025	Membro dell'Editorial Board (Early Career) Sensors — MDPI Dal 18 dicembre 2025.
------	--

2025	Editor del volume APPLEPIES 2025 — Springer LNEE Atti della International Conference on Applications in Electronics Pervading Industry, Environment and Society.
2024	Guest Editor Energies — MDPI Special Issue “Tiny Machine Learning for Energy Applications”, aprile 2024 — agosto 2025.

ORGANIZZAZIONE DI CONVEGNI

Program Chair APPLEPIES 2025 Torino, Italia, 11–12 settembre 2025.
Program Committee APPLEPIES Dal 2023.
Program Committee GALA — Games and Learning Alliance Dal 2023.
Chairman di sessione APPLEPIES 2023 1st Short Oral session — Genova, Italia, 28–29 settembre 2023.

ATTIVITÀ DI REVISIONE

RIVISTE	CONFERENZE
IEEE Accessdal 2023	IEEE ICRA — International Conference on Robotics and Automationdal 2026
Journal of Systems Architecturedal 2023	
Engineering Applications of Artificial Intelligence dal 2024	IEEE/RSJ IROS — Intelligent Robots and Systems dal 2025
Sensors dal 2024	GALA — Games and Learning Alliance dal 2023, 2024
IET Intelligent Transport Systems dal 2025	APPLEPIES dal 2021–2024
IEEE Transactions on Intelligent Vehicles dal 2026	
IEEE Transactions on Circuits and Systems II: Express Briefs dal 2026	
Robotics and Autonomous Systems dal 2026	

Scientific Reports	dal 2026
Journal of Big Data	dal 2026
Discover Internet of Things	dal 2026

08 Interventi a convegni

13 INTERVENTI

11-12 set 2025	APPLEPIES 2025 Torino, Italia
11-14 gen 2025	43rd IEEE ICCE – International Conference on Consumer Electronics Las Vegas, NV, USA
19-20 set 2024	APPLEPIES 2024 Torino, Italia
24-26 giu 2024	55th Annual Meeting of the Italian Society of Electronics (SIE) Genova, Italia
28-29 set 2023	APPLEPIES 2023 Genova, Italia
7-9 set 2023	54th Annual Meeting of the Italian Society of Electronics (SIE) Noto (RG), Italia
26-27 set 2022	APPLEPIES 2022 Genova, Italia
7-9 set 2022	53rd Annual Meeting of the Italian Society of Electronics (SIE) Pizzo (VV), Italia
21-22 set 2021	APPLEPIES 2021 Pisa, Italia
1-2 dic 2021	GALA 2021 – Games and Learning Alliance La Spezia, Italia (online)
7-9 lug 2021	52nd Annual Meeting of the Italian Society of Electronics (SIE) Trieste, Italia (online)
9-10 dic 2020	GALA 2020 – Games and Learning Alliance Laval, Francia (online)
19-20 nov 2020	APPLEPIES 2020 Online

09 Premi e associazioni

2025 **Best Presentation of the Session — sessione “Edge Computing”**
43rd IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 11–14 gennaio 2025
A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, F. Bellotti — “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices”

ASSOCIAZIONI PROFESSIONALI

IEEE
IEEE Member.

IEEE Consumer Technology Society
Membro del Technical Committee “Machine Learning, Deep Learning and AI in CE (MDA)”.

SIE — Società Italiana di Elettronica
Socio nell'Unità di Ricerca dell'Università di Genova dal 2021, area di ricerca “Sistemi elettronici ed applicazioni”.

10 Contatti e profili

EMAIL luca.lazzaroni@unige.it	UNIGE rubrica.unige.it	ORCID 0000-0001-8092-5473
SCOPUS 57220892898	GOOGLE SCHOLAR w_6a6sYAAAAJ	GITHUB @lucalazzaroni
LINKEDIN luca-lazzaroni		

Pubblicazioni e indicatori sono presi da Scopus da un job schedulato, e il PDF viene stampato da questa pagina: i due non possono divergere.

CV completo in PDF. Sorgente del sito su GitHub. Composto in IBM Plex.